Radar Systems Engineering - Exercise Problems

Complete Transcribed Text Collection from Textbook Problem Sets (Chapters 1, 2 & 3)

CHAPTER 1 • An Introduction to Radar

1.1

a. What should be the pulse repetition frequency of a radar in order to achieve a maximum unambiguous range of 60 nmi?

b. How long does it take for the radar signal to travel out and back when the target is at the maximum unambiguous range?

c. If the radar has a pulse width of 1.5 μs, what is the extent (in meters) of the pulse energy in space in the range coordinate?

d. How far apart in range (meters) must two equal-size targets be separated in order to be certain they are completely resolved by a pulse width of 1.5 μs?

e. If the radar has a peak power of 800 kW, what is its average power?

f. What is the duty cycle of this radar?

1.2 A ground-based air-surveillance radar operates at a frequency of 1300 MHz (*L* band). Its maximum range is 200 nmi for the detection of a target with a radar cross section of one square meter (σ = 1 m²). Its antenna is 12 m wide by 4 m high, and the antenna aperture efficiency is ρ_a = 0.65. The receiver minimum detectable signal is S_{min} = 10^{-13} W. Determine the following:

a. Antenna effective aperture A_e (square meters) and antenna gain *G* [numerically and in dB, where *G* (in dB) = 10 $\log_{10} G$ (as a numeric)].

b. Peak transmitter power.

c. Pulse repetition frequency to achieve a maximum unambiguous range of 200 nmi.

d. Average transmitter power, if the pulse width is 2 μs.

e. Duty cycle.

f. Horizontal beamwidth (degrees).

1.3

a. What is the peak power of a radar whose average transmitter power is 200 W, pulse width of 1 μs, and a pulse repetition frequency of 1000 Hz?

b. What is the range (nmi) of this ground-based air-surveillance radar if it has to detect a target with a radar cross section of 2 m² when it operates at a frequency of 2.9 GHz (*S* band), with a rectangular-shaped antenna that is 5 m wide, 2.7 m high, antenna aperture efficiency ρ_a of 0.6, and minimum detectable signal S_{min} equal to 10^{-13} W (based on P_t in the radar equation being the peak power)?

c. Sketch the received echo signal power as a function of range from 10 to 80 nmi.

1.4 The moon as a radar target may be described as follows: average distance to the moon is 3.844×10^8 m (about 208,000 nmi); experimentally measured radar cross section is 6.64×10^{11} m² (mean value over a range of radar frequencies); and its radius is 1.738×10^6 m.

a. What is the round-trip time (seconds) of a radar pulse to the moon and back?

b. What should the pulse repetition frequency (prf) be in order to have no range ambiguities?

c. For the purpose of probing the nature of the moon's surface, a much higher prf could be used than that found in (b). How high could the prf be if the purpose is to observe the echoes from the moon's front half?

d. If an antenna with a diameter of 60 ft and aperture efficiency of 0.6 were used at a frequency of 430 MHz with a receiver having a minimum detectable signal of 1.5×10^{-16} W, what peak power is required? Does your answer surprise you; and if so, why?

e. The radar cross section of a smooth perfectly conducting sphere of radius a is πa^2. What would be the radar cross section of the moon if it were a sphere with a perfectly smooth, conducting surface? Why might the measured cross section of the moon (given above) be different from this value?

1.5 A radar mounted on an automobile is to be used to determine the distance to a vehicle traveling directly in front of it. The radar operates at a frequency of 9375 MHz (*X* band) with a pulse width of 10 ns (10^{-8} s). The maximum range is to be 500 ft.

a. What is the pulse repetition frequency that corresponds to a range of 500 ft?

b. What is the range resolution (meters)?

c. If the antenna beamwidth were 6°, what would be the cross-range resolution (meters) at a range of 500 ft? Do you think this value of cross-range resolution is sufficient?

d. If the antenna dimensions were 1 ft by 1 ft and the antenna efficiency were 0.6, what would be the antenna gain (dB)?

e. Find the average power required to detect a 10 m² radar cross section vehicle at a range of 500 ft, if the minimum detectable signal is 5×10^{-13} W.

1.6 Determine (a) the peak power (watts) and (b) the antenna physical area (m²) which make the cost of the following radar a minimum:

Frequency:	1230 MHz (*L* band)
Antenna aperture efficiency:	0.6
Receiver minimum detectable signal:	3×10^{-13} W
Unit cost of transmitter:	$2.20 per watt of peak power
Unit cost of antenna:	$1400 per square meter of physical area
Cost of receiver and other items:	$1,000,000

The radar must detect a target of 2 m² cross section at a range of 200 nmi.
(You will have to use one of the simple forms of the radar range equation.)

a. What is the cost of the antenna and the cost of the transmitter?

b. In a new radar design, how would you try, as a first attempt, to allocate the costs between the antenna and the transmitter (based only on the answers to the above problem)?

1.7 Who invented radar? (Please explain your answer.)

1.8 Three variants of the simple form of the radar equation have been given. In Eq. (1.9) the wavelength is in the numerator, in Eq. (1.10) the wavelength is in the denominator, and in Eq. (1.7), there is no explicit indication of wavelength. How would you respond to the question: "How does the radar range vary with the radar wavelength, everything else being the same?"

1.9 If the weight of a transmitter is proportional to the transmitter power (i.e., $W_T = k_T P$) and if the weight of an antenna is proportional to its volume (so that we can say its weight is proportional to the 3/2 power of the antenna aperture area A, or $W_A = k_A A^{3/2}$), what is the relationship between the weight of the antenna and the weight of the transmitter that makes the total weight $W = W_T + W_A$ a minimum, assuming a fixed range? (You will need the simple form of the radar equation to obtain a relationship between P and A.)

CHAPTER 2 • The Radar Equation

2.7 A transmission line with loss L is connected to the input of a receiver whose noise figure is F_n. What is the overall noise figure of the combination?

2.8 A radar at a frequency of 1.35 GHz has an antenna of width D = 32 ft, a maximum unambiguous range of 220 nmi, and an antenna scan time (time to make one rotation of the antenna) of 10 s.

a. What is the number of echo pulses per scan received by the radar from a point target? [Use the relationship that the antenna half-power beamwidth in radians is $\theta_B = 1.2\,\lambda / D$ (λ = wavelength).]

b. What is the integration loss and the integration-improvement factor when the probability of detection is 0.9 and the probability of false alarm is 10^{-6}?

2.9 Show that the far right-hand side of Eq. (2.36), a definition of the radar cross section, is the same as the simple radar equation, Eq. (1.6). [It is easier to start with Eq. (2.36) and obtain Eq. (1.6), than vice versa.]

2.10 a. What frequency will result in the maximum radar cross section of a metallic sphere whose diameter is 1 m?

b. At what frequency will the radar cross section of a ball bearing one millimeter in diameter be maximum?

2.11 a. What is the maximum radar cross section (square meters) of an automobile license plate that is 12 inches wide by 6 inches high, at a frequency of 10.525 GHz (the frequency of an *X*-band speed radar)?

b. How many degrees in the vertical plane should the plate be tilted in order to reduce its cross section by 10 dB? For purposes of this problem you may assume the license plate is perfectly flat. The radar cross section of a flat plate as a function of the incidence angle ϕ may be written, for ϕ not too large, as:

$$\sigma(\phi) = \sigma_{max}\,[\,\sin^2(2\pi\,(H/\lambda)\sin\phi) / (2\pi\,(H/\lambda)\sin\phi)^2\,]$$

where σ_{max} = maximum radar cross section of a flat plate = $4\pi A^2 / \lambda^2$, A = area of plate, λ = radar wavelength, and H = height of the plate. (Be careful of units. You will have to sketch a portion of the cross section pattern as a function of ϕ to find the value of ϕ corresponding to -10 dB.)

c. What other parts of an automobile might contribute to its radar cross section when viewed directly from the front?

2.12 Describe briefly the behavior of the radar cross section (in the microwave region) of a raindrop and a large aircraft with respect to its dependence on (a) frequency and (b) viewing aspect.

2.13 Describe the chief characteristics of the radar echo from a target when its radar cross section is in the (a) Rayleigh region, (b) resonance region, and (c) optical region.

2.14 A typical value of an individual "sea spike" echo at *X* band (wavelength = 3.2 cm), as discussed in Sec. 7.4, might be 1 m^2. What is the dimension of the side of a square flat plate that produces the same radar echo when the plate is viewed at normal incidence?

2.15 A radar noncoherently integrates 18 pulses, each of uniform amplitude (the nonfluctuating case). The IF bandwidth is 100 kHz.

a. If the average time between false alarms is 20 min, what must be the signal-to-noise ratio per pulse $(S/N)_1$ in order to achieve a probability of detection of 0.80? (Suggest the use of Albersheim's equation.)

b. What is the corresponding value of $(S/N)_n$?

c. What would $(S/N)_1$ be if the target cross section fluctuated according to a Swerling Case 1 model?

2.16 Why does the cross section of a complex target, such as that in Fig. 2.15, fluctuate so rapidly with a small change in aspect angle when the radar wavelength is small compared to the target's dimensions?

2.17 Show that the probability density function for the Swerling Case 1 model is the same as the chi-square of degree 2 [Eq. (2.47)].

2.18
a. What signal-to-noise ratio is required for a radar that makes a detection on the basis of a single pulse, when the probability of detection is 0.50 and the probability of false alarm is 10^{-6}? Assume a nonfluctuating target echo.
b. Repeat for a 0.99 probability of detection and the same probability of false alarm.
c. Repeat parts (a) and (b), but for a Swerling Case 1 fluctuating target.
d. Compare your results in a table. What conclusions can you obtain from this?

2.19 A radar measures an apparent range of 7 nmi when the prf is 4000 Hz, but it measures an apparent range of about 18.6 nmi when the prf is 3500 Hz. What is the true range (nmi)?

2.20
a. Show that the echo signal power P_r received from an aircraft flying at a constant height h over a perfectly conducting flat earth is independent of the range R, when the antenna elevation gain varies as the cosecant-squared of the elevation angle ϕ (that is, $G = G_0 \csc^2 \phi$).
b. In addition to having a received signal that is independent of the range (requiring less dynamic range in the receiver), what is another reason for employing an antenna with a $\csc^2 \phi$ elevation pattern for an air-surveillance radar when compared to a conventional unshaped fan-beam elevation pattern?
c. What are the limitations in applying the simple result of (a) to a radar in the real world?

2.21 A surface-based air-surveillance radar with a fan-beam antenna that rotates 360° in azimuth has a maximum range of 150 nmi and height coverage to 60,000 ft. Its maximum elevation-angle coverage is 30°. What percentage of the total available volume coverage is lost because of the overhead "hole" compared to a radar with complete angular coverage up to 90° (no hole in the coverage)? Assume, for simplicity, a flat earth.

2.22 A radar receives five pulses within its half-power (3 dB) beamwidth as the antenna beam scans past a point target. The middle of the five pulses is transmitted when the maximum of the antenna pattern points in the direction of the target. The first and the fifth pulses are transmitted when the leading and trailing half-power points are, respectively, directed at the target. What is the two-way beam-shape loss (dB) in this case?

2.23 Five identical radars, each with a receiver having a square-law detector, have partial overlap in their radar coverages so that not all radars are guaranteed to see each target. The outputs of all five radars are combined before a detection decision is made. If a target is seen on only one of the five radars and the other four radars see only receiver noise, what is the collapsing loss when the detection probability is 0.5 and the false-alarm probability is 10^{-6}?

2.24 A civil marine radar is employed on boats and ships for observing navigation buoys, detecting land-sea boundaries, piloting, and avoiding collisions. Consider the following civil marine radar:

Frequency:	9400 MHz (*X* band)
Antenna horizontal beamwidth:	0.8°
Vertical beamwidth:	15°
Gain:	33 dB
Azimuth rotation rate:	20 rpm
Peak power:	25 kW
Pulse width:	0.15 μs
Pulse repetition rate:	4000 Hz
Receiver noise figure:	5 dB
Receiver bandwidth:	15 MHz
System losses:	12 dB
Average time between false alarms:	4 hours

a. Plot the single-scan probability of detection as function of range (nmi), assuming a constant cross-section target of 10 m² (a navigation buoy) and free-space propagation. [You will find it easier to select the probability of detection and find the corresponding signal-to-noise ratio; rather than the reverse. You need only consider probabilities of detection from 0.30 to 0.99. You may, for purposes of this problem, select a single (average) value of the integration improvement factor rather than try to find it as a function of P_d (since the curve in the text does not permit otherwise).]

b. Repeat (a) for a Swerling Case 1 target fluctuation model with average cross section of 10 m². Plot on the same diagram as (a).

c. Comment on whether the average power of this radar is too low, just right, or too high for the job it has to perform here.

d. Why do you think this ship-mounted radar antenna has a 15° elevation beamwidth when all the targets are located on the surface of the sea?

2.25 Consider the following air-surveillance radar:

Frequency:	2.8 GHz (*S* band)
Peak power:	1.4 MW
Pulse width:	0.6 μs
Pulse repetition frequency:	1040 Hz
Receiver noise figure:	4 dB
Antenna rotation rate:	12.8 rpm
Antenna gain:	33 dB
Antenna azimuth beamwidth:	1.35 deg
System losses:	12 dB
Average false-alarm time:	20 min
Target cross section:	2 m²

Plot each of the following on the same coordinates (with range as the abscissa):

a. The free-space single-scan probability of detection as a function of range (in nautical miles) for a constant cross-section target. [You will find it easier to select the probability of detection and find the corresponding signal-to-noise ratio, rather than the reverse. You need only consider probabilities of detection from 0.30 to 0.99. You may, for purposes of this problem, select a single (average) value of the integration improvement factor rather than try to find it as a function of P_d (since the curve in the text does not permit otherwise).]

b. The probability of detection as a function of range for the same situation as part (a) but with the detection criterion that the target must be found on at least 2 out of 3 scans of the rotating antenna. [You may assume that the range and the received signal power do not change appreciably over the three scans. For convenience of this calculation, you may assume that the single-scan false-alarm probability is the same as used in part (a).]

c. Repeat (a) for a Swerling Case 1 with average target cross section of 2 m².

d. Repeat (b) for a Swerling Case 1 with average target cross section of 2 m².

e. Is the prf adequate for avoiding range ambiguities?

> *(The radar in this problem is similar to the airport surveillance radar known as the ASR.)*

2.26 Starting with Eq. (2.51) derive the surveillance radar equation (Eq. (2.63)). You may omit the propagation factor, attenuation, and fluctuation loss.

2.27 What is the effect of receiver bandwidth on the maximum range of a well-designed radar, assuming the average power remains constant? Explain your answer.

2.28

a. What is the probability of detecting a target on at least 2 out of 4 scans when the single-scan probability of detection is 0.8?

b. What is the corresponding probability of false alarm in this case when the single-scan false-alarm probability is 10^{-5}?

c. What should be the single-scan false-alarm probability if the overall false-alarm probability with a detection criterion of 2 out of 4 scans is 10^{-5}?

d. When the higher single-scan probability of false alarm of part (c) is employed rather than a 10^{-5} single-scan probability of false alarm, what reduction in the signal-to-noise ratio can be obtained?

2.29 In this problem, it is assumed that the targets for an air-surveillance radar are characterized by a Swerling Case 1 model. There are n pulses received from a target. Half of the n pulses are at one frequency and the other half are at a second frequency that is far enough removed from the first to completely decorrelate the second set of $n/2$ pulses relative to the first set of $n/2$ pulses. What is the improvement in signal-to-noise ratio obtained from this use of frequency diversity when (a) $P_d = 0.95$ and (b) $P_d = 0.67$? (c) If the radial extent (in range) of a target is 30 m, what must the difference in the two frequencies be to decorrelate the target echo?

2.30

a. Make a list of the system losses that might occur in a long-range air-surveillance radar, and estimate an approximate value for the loss due to each factor. You need not include losses due to doppler processing. (There is, of course, no unique answer for this question.)

b. Using the total system loss you have estimated, what is the reduction in radar range that occurs due to the system losses if the radar range without losses is 200 nmi?

2.31 Question 1.8 of Chapter 1 asked "How does radar range depend on the wavelength?" Based on Chapter 2, how would you now answer this question for an air-surveillance radar? (Please justify your answer.)

2.32 An experimental measurement of the blip-scan ratio (single-scan probability of detection) of a particular target at a particular range gives a value of 0.5 after 10 trials (antenna scans).

a. What is the confidence interval of this measurement if the confidence level has to be 90 percent?

b. What is the confidence interval (with the same confidence level of 90 percent) after 100 scans, assuming the measured blip-scan ratio is still 0.5?

CHAPTER 3 • MTI and Pulse Doppler Radar

3.1 A satellite orbiting the earth in a circular orbit at an altitude of 5000 nmi has a speed of 2.7 nmi/s.

a. What is the doppler frequency shift if the satellite is observed by a UHF radar (450 MHz) lying in the plane of orbit, just as the satellite appears over the horizon? (The radius of the earth is 3440 nmi. You may ignore the effects of refraction by the earth's atmosphere and reflection from the earth's surface.)

b. What is the doppler frequency shift when the satellite is observed at the zenith?

3.2 A VHF radar at 220 MHz has a maximum unambiguous range of 180 nmi.

a. What is its first blind speed (in knots)?

b. Repeat, but for an *L*-band radar at 1250 MHz.

c. Repeat, but for an *X*-band radar at 9375 MHz.

d. What would be the unambiguous range (nmi) of the *X*-band radar of part (c) in order to give the same blind speed you found in part (a) for the VHF radar?

e. If you needed to have a radar with the first blind speed of the VHF radar of part (a), would you rather have the VHF radar of part (a) or the *X*-band radar of part (d)? Please explain your answer. (There might not be a unique answer.)

3.3 An *L*-band radar (1250 MHz) has a prf of 340 Hz. It detects a rainstorm moving at a radial velocity of 12 kt. Assume that the width of the thunderstorm's doppler spectrum is very small (a narrow spectral line; which is not, of course, reality but it makes the problem easier). The radar employs a single delay-line canceler.

a. How much does the single delay-line canceler attenuate (in dB) the storm echo compared to the response if the storm were moving with a radial velocity corresponding to the velocity which gives the maximum filter response?

b. What would be the attenuation of the storm compared to the maximum response, if a double delay-line canceler were used?

3.4 a. Show that the product of the maximum unambiguous range R_{un} and the first blind speed v_1 is equal to $c\,\lambda / 4$, where c = velocity of propagation and λ = radar wavelength.

b. What guidance, if any, does this relation give for avoiding ambiguities?

3.5 What is the highest frequency that a radar can be operated if it is required to have a maximum unambiguous range of 200 nmi and no blind speeds less than 600 kt?

3.6 Show that a triple delay-line canceler is equivalent to a four-pulse delay-line canceler with weights equal to the coefficients of the binomial expansion with alternating sign.

3.7 a. Derive the expression for the ratio v_1 / v_{av}, where v_1 is the first blind speed of a staggered prf with N different prfs, and v_{av} is the first blind speed obtained with a constant prf waveform equal to the average of the N staggered prfs.

b. What is the ratio v_1 / v_{av} when $N = 4$ and the prfs are related as 30:35:32:36?

3.8 a. How can the transmission of N constant-prf radar waveforms, each at a different RF frequency, be used to avoid blind speeds?

b. Derive an expression for v_1 / v_{av}, where v_1 is the first blind speed when N different RF frequencies are transmitted, all at the same prf, and v_{av} is the blind speed when only a single RF frequency is transmitted equal to the average of the N RF-frequencies.

c. Is there any advantage in changing both the prf and the RF frequency to avoid blind speeds?

3.9 An *S*-band (3.1 GHz) air-surveillance radar utilizes a staggered waveform with four different prfs, which are 1222, 1031, 1138, and 1000 Hz.

a. What is the first blind speed (knots) if a constant prf is used which has a pulse repetition period equal to the average of the four periods of the staggered waveform?

b. What is the first blind speed (knots) of the staggered prf waveform? Note that the n_i for these four frequencies are 27, 32, 29, 33 respectively.

c. What is the maximum unambiguous range of the staggered prf waveform?

d. What is the depth (dB) of the first null of the staggered prf waveform?

e. What is the maximum MTI improvement factor for the staggered prf waveform, assuming a gaussian clutter spectrum with a standard deviation of 10 Hz?

3.10 a. What is the first blind speed (knots) of an *L*-band radar (1250 MHz) when the prf has a maximum unambiguous range of 240 nmi?

b. Determine the periods of a pulse-to-pulse staggered MTI waveform with three different periods, for the purpose of increasing the radar's first blind speed found in (a) to a value no less than 1200 kt. The maximum unambiguous range of the three periods is to be no less than 240 nmi. (Note: There is no unique answer for this part. In practice the choice of the three periods also should be selected to achieve an acceptable null depth and a desired improvement factor, something beyond the scope of this particular problem.)

3.11 a. In a digital filter bank with 16 filters, what phase increment (degrees) is required for the phase shifts at each of the 16 taps of the delay line so as to generate the filter that is adjacent to the zero doppler filter?

b. What is the null-width (the distance between the two nulls that define the main response) of the filters if the pulse repetition frequency of the radar is 2560 Hz?

c. If a four-pulse canceler precedes the 16-tap delay line of the filter bank, how many pulses are there in a coherent processing interval?

3.12 Consider an MTD processor with a bank of eight contiguous doppler filters installed in an airport surveillance radar operating at *S* band (2.8 GHz). The MTD uses two different pulse repetition frequencies to unmask a moving target whose echo signal might be in the same doppler filter as moving weather clutter. Let one of the two prfs be 1100 Hz. Weather clutter is assumed to have a spectrum of radial velocities extending from 0 to 25 kt. An aircraft flying at a radial velocity of 250 kt is aliased in doppler and falls into the same doppler filter (near the middle of filter no. 2—please verify this) as the weather clutter. It is masked by the clutter and not detected. (Note that filter no. 1 is designated as the filter centered at zero frequency.)

a. What should be the second prf (smaller than the first, so as not to decrease the maximum unambiguous range) in order to move the aliased aircraft velocity completely outside of the main response of filter no. 2 and into the middle of filter no. 4?

b. What percent change has been made in the first prf to arrive at the second prf?

c. Instead of changing the prf to unmask the target, how much should the RF frequency be changed in order to unmask the target (by shifting it to the middle of the fourth filter) when the prf is kept at 1100 Hz?

3.13 An MTI radar with a single delay-line canceler has the following characteristics:

Frequency:	3000 MHz
Azimuth beamwidth:	1.2 degrees
Antenna rotation rate:	10 rpm
Prf:	1000 Hz
A/D converter:	8 bits quantization
Stalo phase stability:	0.6 degree
Clutter standard deviation σ_v:	0.3 m/s

a. Determine which of the following is the major limitation to the overall MTI improvement factor for this radar:
   - stability of the stalo (due to phase changes)
   - clutter internal motion
   - antenna scanning modulation
   - A/D converter noise

b. What is the overall improvement factor (dB) for this radar?

3.14 A radar is to have a total improvement factor of 45 dB. Its frequency is 3.0 GHz, the prf is 340 Hz, and it uses a single delay-line canceler. Assume that the four factors asked for below in (a) through (d) are independent of one another and that their contributions to the overall improvement factor are allocated equally.

a. What must be the overall phase stability (degrees)?

b. What must be the relative amplitude stability (percent)?

c. What is the minimum number of bits required for the A/D converter?

d. What is the maximum permissible rms velocity spread (m/s) of the clutter fluctuations?

e. What should be the limit level of the receiver?

3.15

a. What is the limitation to the improvement factor due to antenna scanning modulation (finite time on target) with a radar that has a beamwidth of 1.5 degrees, prf of 340 Hz, and antenna scan rate of 6 rpm when a single delay-line canceler is used?

b. What two basic things can the radar systems designer consider to increase to at least 40 dB the limitation to the improvement factor caused by antenna scanning?

c. Which of your two methods in part (b) do you think is the better option?

3.16 In a rotating reflector antenna with dimension D, what is the relationship between the spread in doppler frequency due to the finite time on target and the doppler frequency shift from the tip (end) of the rotating antenna? (Assume that the beamwidth of the antenna is $\theta_B = \lambda / D$ radians.)

3.17

a. Assume the MTI improvement factor is determined only by the internal motion of the clutter. How much will the improvement factor be decreased (in dB) if the frequency of a radar using a three-pulse canceler were changed from 430 MHz (UHF) to 3.3 GHz (*S* band)?

b. How much should the prf of the *S*-band radar be increased to make its improvement factor equal that of the UHF radar?

3.18 If internal motion of the clutter caused by the wind were the only factor affecting the clutter spectrum, what would be the improvement factor at a frequency of 10 GHz for a breezy wind (9 kt), prf of 1000 Hz, and a double delay-line canceler, when the clutter spectrum is modeled by an exponential clutter spectrum? (Be careful of units.)

3.19 Show that the spread in the AMTI radar clutter spectrum due to movement of the radar (platform motion), which is given by Eq. (3.58) as $\Delta f_c = (2\, v / \lambda)\, \theta_B \sin\theta$, is independent of frequency, assuming both the velocity v and the antenna dimension D are constant.

3.20 Assuming an airborne air-surveillance (AMTI) radar flying at a speed of 300 kt with a rotating fan-beam antenna whose horizontal dimension D = 24 ft, plot the doppler shift f_c of the ground clutter echo and the spread in doppler shift Δf_c as a function of the azimuth angle (from 0 to 180 degrees) for frequencies of 420 MHz and 3.5 GHz. (OK to assume the elevation angle is zero in this problem and that the azimuth beamwidth in degrees is $\theta_B = 65\ \lambda / D$, where λ is the wavelength.)

3.21 Consider an AMTI radar with a frequency of 440 MHz flying in an aircraft at a speed of 320 kt. Its azimuth beamwidth is 6 degrees and its prf is 330 Hz.

a. What is the doppler frequency of the clutter echo and the spread in the clutter doppler at azimuth angles of 0, 45, and 90 degrees, where 0 is head-on and 90 is broadside? (You may assume that the elevation angle is zero, which is unrealistic of course, but it makes the problem simpler.)

b. Assume that TACCAR is applied so that the doppler clutter frequency is fully compensated along the center of the main beam (that is, the center frequency of the clutter doppler spectrum is at zero doppler frequency). DPCA is not applied. Sketch the doppler space (the resulting clutter spectrum as a function of doppler frequency) for the case where the radar antenna is pointing broadside at 90 degrees azimuth angle. For this problem you may assume that σ_c is the same as $\Delta f_c / 2$. (Draw approximately to scale along the frequency axis.)

c. What is the value of σ_c / f_p when the antenna is pointing broadside as it is in part (b), where f_p = prf and σ_c = standard deviation of the clutter spectrum which can be approximated here by $\Delta f_c / 2$?

d. How well do you think a radar of this type detects moving aircraft targets in clutter?

3.22 a. Why does a high-prf pulse doppler radar require a much larger improvement factor than a low-prf MTI radar, assuming comparable performance in detecting moving targets in clutter?

b. Why does a high-prf pulse doppler radar (such as AWACS) generally need more average power than an AMTI radar of comparable performance?

c. Why does a high-prf pulse doppler radar not need DPCA as does an AMTI radar?

3.23 a. What does a medium-prf pulse doppler radar do better than a high-prf pulse doppler radar?

b. What does a high-prf radar do better than a medium-prf pulse doppler radar?

3.24 Why can't the altitude line in a high-prf pulse doppler radar be eliminated by range gating rather than by filtering?

3.25 An HF over-the-horizon radar for detection of commercial aircraft targets out to a range of 2000 nmi might, for example, operate at a frequency of 15 MHz and a prf of 30 Hz. It employs doppler processing to separate moving targets from clutter. Is it an MTI radar, a pulse doppler radar, or what? Explain your answer.